

Northwind · paid media review (simulated store)

Jul 27-Aug 9, 2026 Compared with Jul 13-26, 2026 1 account · USD

This is a simulated account, so results are scored against the simulation's truth, not real sales. Over 8 weeks, the agent's weekly budget moves produced 39.6 conversions a day against 37.3 with budgets left alone. That is +6.1% at the same total budget, and it captured 86% of the available gain. TabPFN raised 4 alerts on unusual days and 3 were real problems. The $\pm 50\%$ day-over-day rule raised 24 alerts and 22 were false alarms. Next, move budget toward Retargeting and investigate the flagged days.

SPEND	CONVERSIONS	CPA	ROAS
30,077.93 USD	468	64.27 USD	1.21
-0.0% vs previous	-21.1% vs previous	+26.7% vs previous	-22.8% vs previous

01 What was unusual

TabPFN, 95% expected range · recorded answers

TabPFN raised 4 alerts; 3 were real problems. A $\pm 50\%$ day-over-day rule raised 24 for the same days, 22 of them false alarms.

How TabPFN is applied: one real day, step by step

1 WHAT GOES IN

A table with one row per campaign per day: 320 rows of history with the spend known, and 56 rows to judge. Three of them:

CAMPAIGN	DAY	USUAL OF LAST 7 DAYS	A WEEK BEFORE	BUDGET	SPEND
Retargeting	Fri, Jul 24	282	250	325	302
Retargeting	Sat, Jul 25	287	287	325	319
Retargeting	Wed, Aug 5	227	244	183	?

Each row also carries the platform; the day number, for trend; a simple estimate: the usual level scaled by the budget change. One request per metric: 2 for this page.

2 WHAT COMES OUT

For each row to judge, a range: where 95 in 100 normal days would land. For Retargeting on Aug 5:

spent 301

expected 170

134

206

LOW (2.5%)	EXPECTED	HIGH (97.5%)
134.29	169.61	205.92

3 THE VERDICT

Retargeting spent 300.59 USD, above the top of the range: an alert. In the simulation this day was: **spend spike, caught.**

WHY A RANGE PER ROW BEATS A FIXED RULE

- **The range moves with the budget.** On Jul 27 Retargeting's budget went from 325 to 244, and the expected spend went from 299 to 224 with it. No alert: the budget is in the row, so a deliberate change is not mistaken for a problem.
- **The range is as wide as the campaign is noisy.** Retargeting went from 1 to 5 conversions on Aug 2 (+400%), so the $\pm 50\%$ rule alerts. The model's range for that day was 0.6 to 12.6: no alert, and nothing was planted there. The rule raised 19 more alerts like it.
- **Nothing to train or tune.** TabPFN is a model pretrained on synthetic tables. The history rows and the rows to judge go to it in one request and the ranges come back; nothing is trained or tuned for this account.

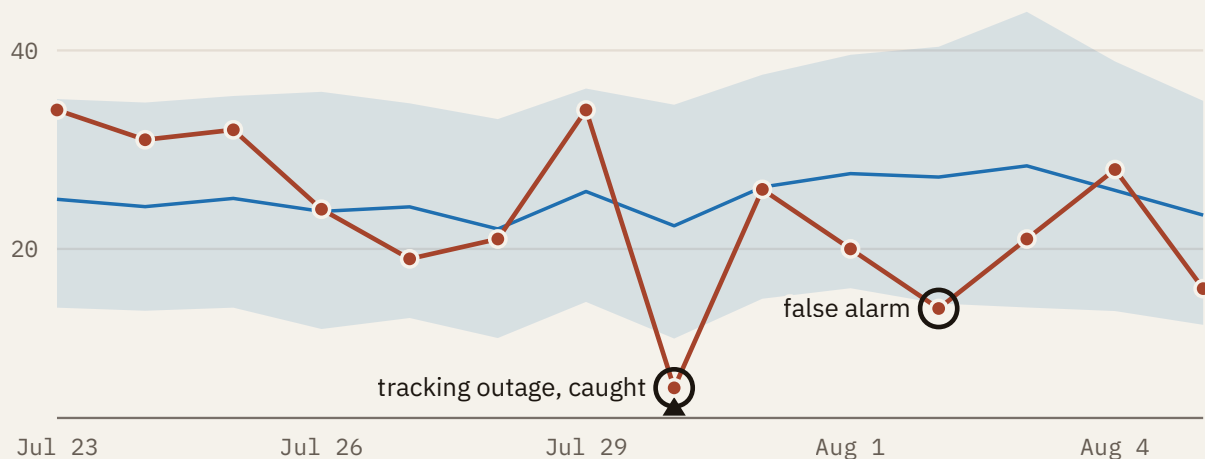
THE RESULT, ON THE SAME DAYS

TabPFN, 95% expected range		3 real of 3 · 1 false alarm
Local model, 95% expected range		3 real of 3 · 5 false alarms
$\pm 50\%$ day-over-day rule		2 real of 3 · 22 false alarms

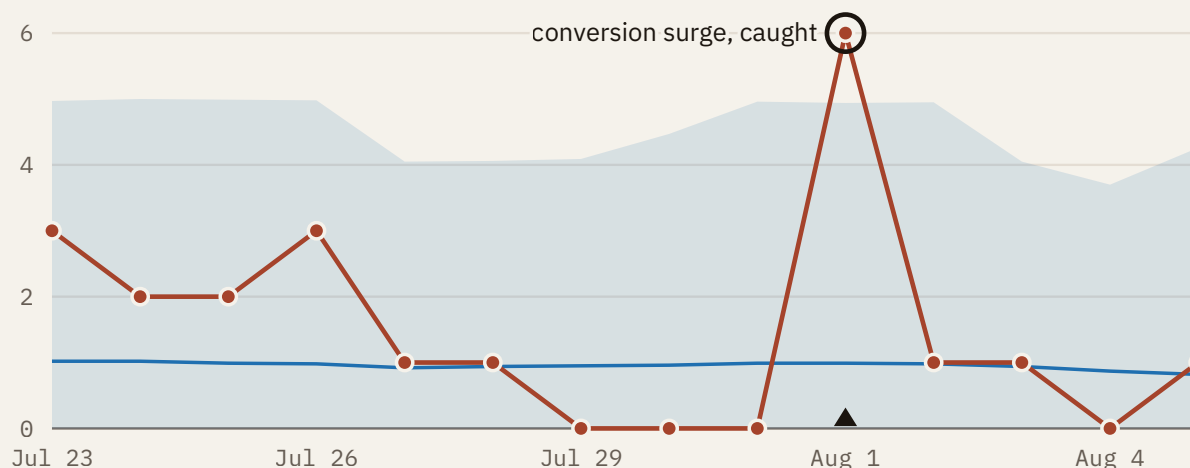
Real problem caught False alarm

Observed Expected, with its range Flagged Planted in the simulation

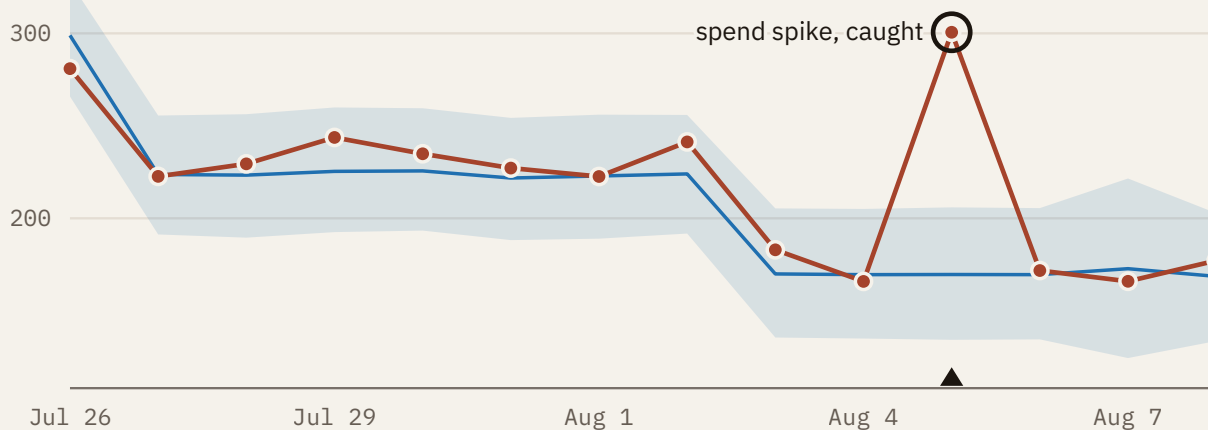
SHOPPING · CONVERSIONS



BRAND SEARCH · CONVERSIONS



RETARGETING · SPEND · USD



Spend checked Jul 26-Aug 8, 2026; Conversions checked Jul 23-Aug 5, 2026. Conversions are checked on earlier days because recent ones are still arriving. 5 campaign charts with nothing flagged or planted are not shown. These are the model's answers recorded earlier for this same account and replayed; no call was made.

Across three simulated accounts of about 450 campaign-days each: TabPFN's 95% range had precision 0.34 and 21 false alarms per account; the local model 0.24 and 33; the $\pm 50\%$ day-over-day rule 0.07 and 169.

02 What the budget moves bought

TabPFN as the global model · recorded answers

Over 8 weeks the agent's weekly reallocations produced 39.6 conversions a day, against 37.3 with the budgets left alone: +6.1% at the same total budget. The best possible split would have produced 40.0, so the agent captured 86% of the gain there was to have.

Each week the model predicts how many conversions each campaign would get at budgets it has not tried, and the agent moves budget to where the next unit of spend buys the most, at most 25% per campaign per week. All three lines are scored by the simulation's true curves, not by any fitted model.

BUDGETS LEFT ALONE

37.3 conversions a day
the comparison

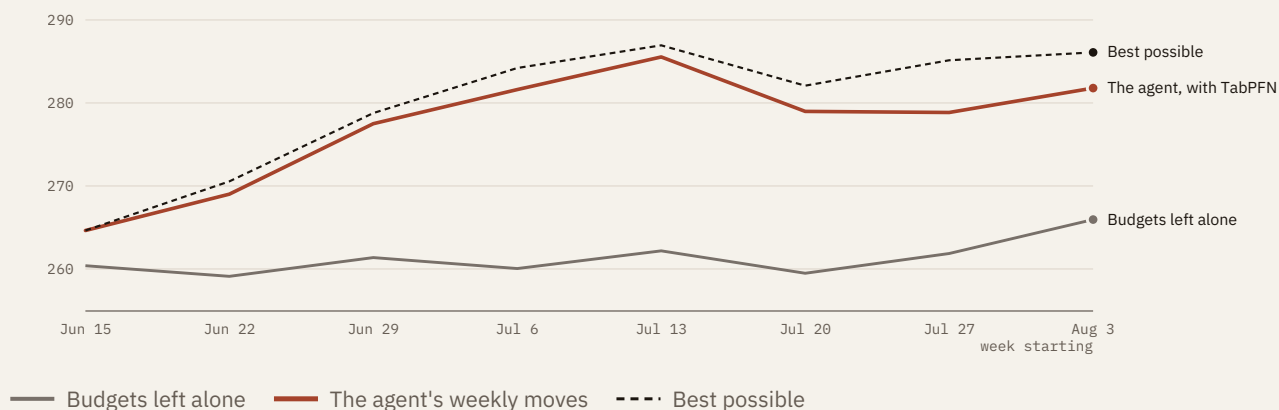
THE AGENT, WITH TABPFN

39.6 conversions a day
+6.1% against budgets left alone

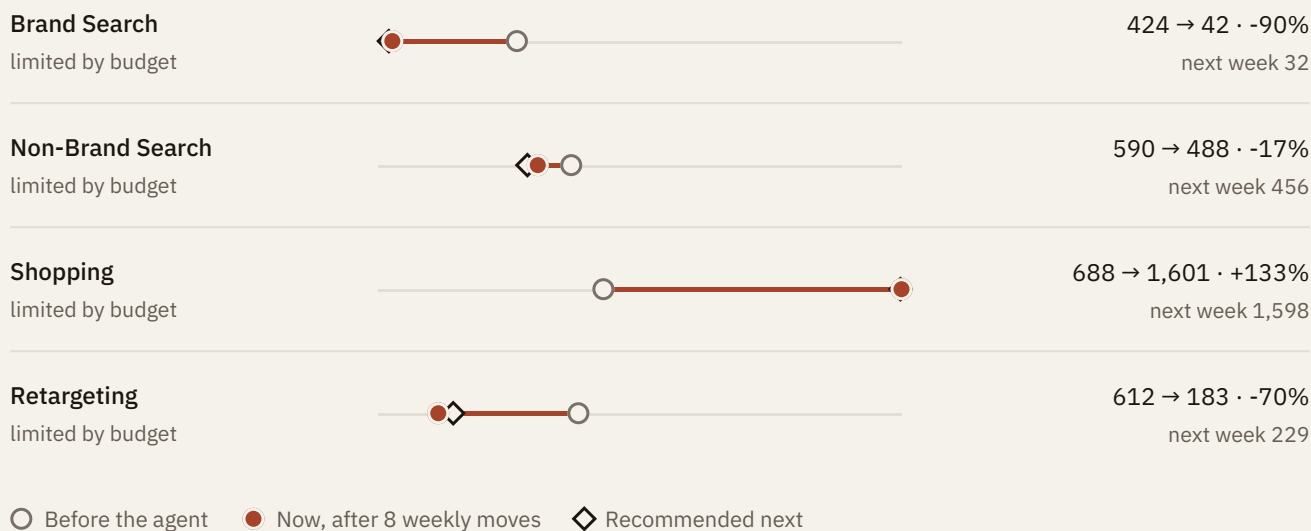
BEST POSSIBLE

40.0 conversions a day
+7.1%, knowing the true curves

CONVERSIONS PER WEEK



DAILY BUDGET BY CAMPAIGN · USD



How TabPFN is applied: one real campaign, step by step

1 WHAT GOES IN

A table with one row per campaign per day, from all 4 campaigns together: 348 rows with the conversions known. Each row carries the campaign's cost per conversion, the weekday, and the spend.

2 WHAT IT IS ASKED, AND ANSWERS

What would each campaign get at 64 spend levels, averaged over weekdays? For Shopping, two of its days and two of the questions:

CAMPAIGN	DAY	SPEND	CONVERSIONS
Shopping	Sun, Aug 2	1,341	14
Shopping	Mon, Aug 3	1,475	21
Shopping	any weekday	1,006	? → 19.5 predicted
Shopping	any weekday	1,677	? → 29.9 predicted

The answers join the campaign's own days when its spend curve is fitted, so the curve is not drawn from that campaign's few days alone.

3 WHAT THE PREDICTIONS DECIDE

Budget goes to where the next 100 a day buys the most. Before the agent that was Shopping: 1.81 more conversions a day by the fitted curve, 1.78 by the true one.

NEXT 100 A DAY BUYS	BEFORE THE AGENT (TRUE)	NOW (TRUE)	SO FAR	NEXT
Brand Search	0.81 (1.09)	1.19 (1.43)	↓ cut	↓ cut
Non-Brand Search	1.50 (1.07)	1.55 (1.15)	↓ cut	↓ cut
Shopping	1.81 (1.78)	1.57 (1.53)	↑ raised	keep
Retargeting	1.42 (1.27)	1.75 (1.63)	↓ cut	↑ raise

Extra conversions a day, by the curves fitted today, with the simulation's true figure in brackets. A split is balanced when the next unit buys about the same everywhere. The gap between the best and the worst campaign went from 1.00 to 0.56 (by the true curves, from 0.71 to 0.48).

WHY PREDICT AT ALL

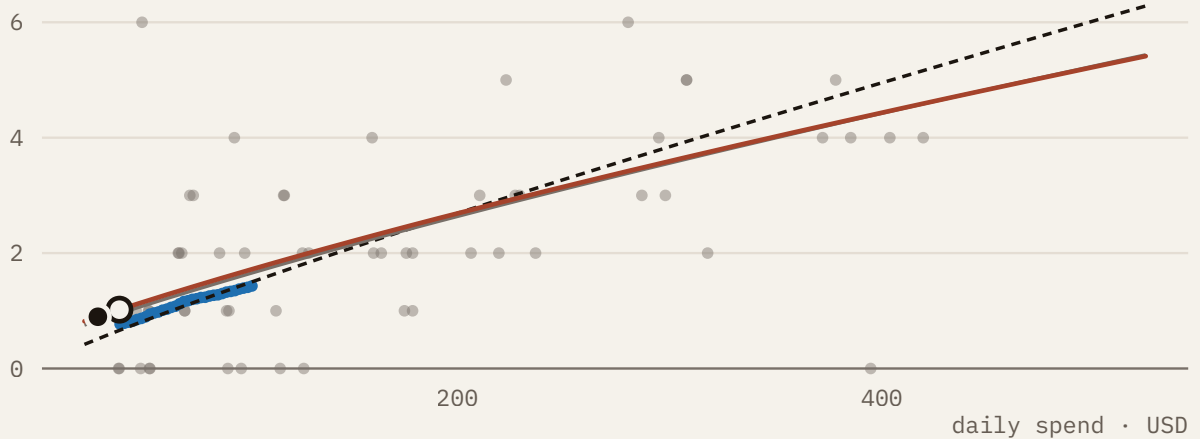
- **A campaign's own days do not reach far enough.** Shopping had never spent more than 1,122 a day before the agent; its budget is now 1,601. Its own history could not say what that would buy. The model sees every campaign's days at once, so each borrows the response seen in the others.
- **Nothing to train or tune.** TabPFN is a model pretrained on synthetic tables. Every campaign's days and the questions go to it in one request and the predictions come back; it assumes no curve shape and nothing is trained for this account.

Each campaign's spend response curve

Conversions a day against daily spend. Each curve is fitted to the campaign's own days and to points the global model predicts from every campaign's history, at spends the campaign has not tried. These are the model's answers recorded earlier for this same account and replayed; no call was made.

● A day of history ● Predicted by the global model — Fitted curve — Fitted with the pooled regression
- - - True curve (simulation)

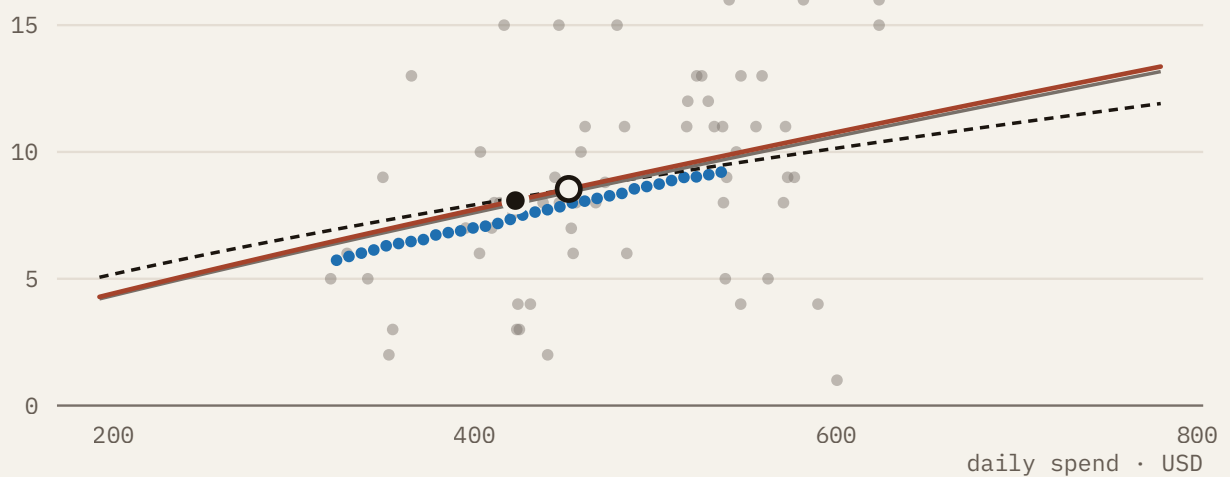
BRAND SEARCH



○ Now 42

● Recommended 32

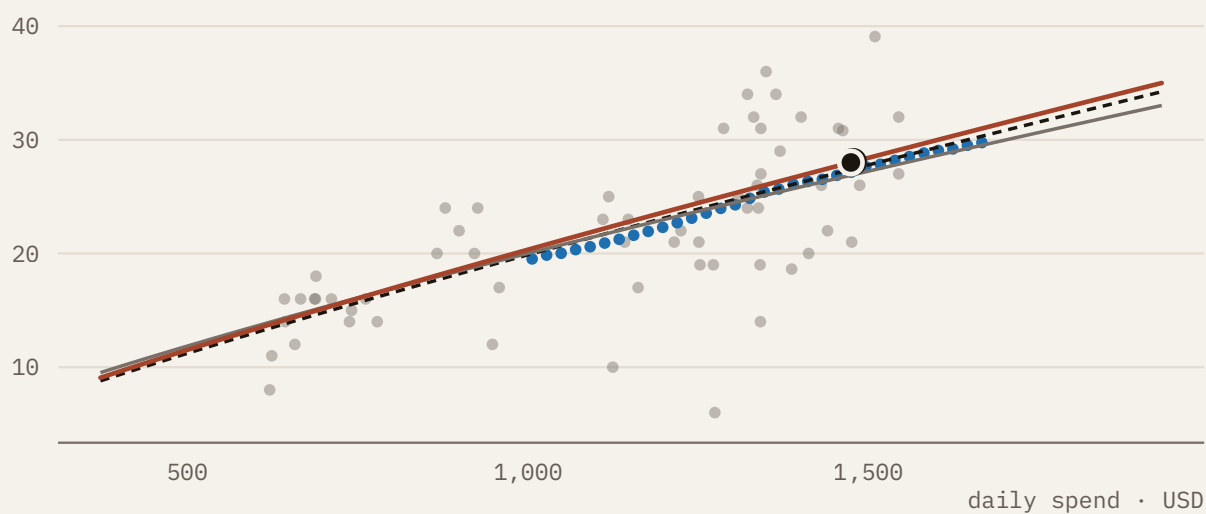
NON-BRAND SEARCH



○ Now 488

● Recommended 456

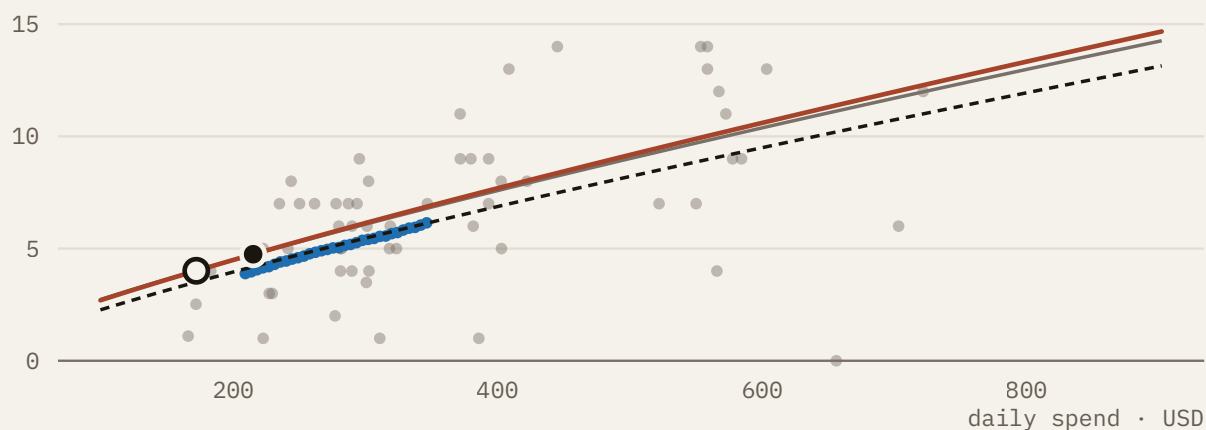
SHOPPING



○ Now 1,601

● Recommended 1,598

RETARGETING



○ Now 183

● Recommended 229

CAMPAIGN	SPEND LIMITED BY	BUDGET NOW	RECOMMENDED	CONVERSIONS A DAY	ELASTICITY (TRUE)
Brand Search	budget	42.43	31.82	1.0 → 0.9	0.72 (0.88)
Non-Brand Search	budget	487.65	455.78	8.5 → 8.1	0.82 (0.61)
Shopping	budget	1,600.90	1,597.64	28.1 → 28.0	0.82 (0.83)
Retargeting	budget	182.96	228.70	4.0 → 4.7	0.79 (0.80)

Budgets total 2,313.94 now and 2,313.94 as recommended (USD a day). Global model: tabpfn:v3.5 (recorded). Nothing has been changed.

In this project's tests across simulated accounts, the simpler pooled regression has so far been the better budget predictor than TabPFN, which is why it is the default; both do far better than leaving budgets alone. The grey curve is the pooled fit, for comparison.

03 Next steps

Written by the agent (anthropic/claude-sonnet-5.5), recorded

RETARGETING DAILY BUDGET

Medium confidence

Raise the daily budget from 182.96 USD to the recommended 228.70 USD, an increase of 45.74 USD (25.0%). Submit it as a proposal for approval, funded by the cuts below so the total stays at 2313.94 USD.

Retargeting is limited by budget and is the only campaign expected to gain. Expected conversions rise from 4.01 to 4.75 a day, about 0.74 more. The 8-week trial of this approach gained 6.1% at the same total budget.

Measurement & follow-up

Expected effect

About 0.74 more conversions a day from Retargeting. This is a model estimate, so the real gain may differ.

Measurement

Compare Retargeting conversions a day and total account conversions a day against the trial level of 39.6 over the next few weeks. Watch Retargeting spend for spikes, since one was flagged there on 2026-08-05.

Reversal

Set the Retargeting budget back to 182.96 USD through a new proposal, and restore the funding campaigns to their current budgets.

BRAND SEARCH AND NON- BRAND SEARCH DAILY BUDGETS

Low confidence

Lower Brand Search from 42.43 USD to 31.82 USD (-10.61 USD) and Non-Brand Search from 487.65 USD to 455.78 USD (-31.87 USD). Leave Shopping almost flat, from 1600.90 USD to 1,597.64 USD. Submit these as proposals together with the Retargeting increase.

The result recommends these shifts at no change in total budget. Expected conversions fall by 0.46 a day for Non-Brand Search (8.54 to 8.08) and by 0.12 for Brand Search (1.02 to 0.90). Together that is 0.58 a day lost. Shopping is expected to lose 0.05 a day.

Measurement & follow-up

Expected effect

The expected net gain across all four campaigns is small, about 0.11 conversions a day. The Retargeting gain only slightly outweighs the losses elsewhere, so the benefit depends on the model being right.

Measurement

Track conversions a day for each campaign over the next few weeks and compare them with the expected values above. Confirm the account total does not fall below the current level.

Reversal

Restore Brand Search to 42.43 USD and Non-Brand Search to 487.65 USD through a new proposal if account conversions drop.

UNUSUAL DAYS: SHOPPING TRACKING, RETARGETING SPEND, BRAND SEARCH CONVERSIONS

High confidence

Check the Shopping conversion tracking for 2026-07-30. Check the Retargeting spend settings and change history for 2026-08-05. Review the Brand Search conversion surge on 2026-08-01 to confirm it was genuine. Keep the TabPFN 95% expected range as the alert method, and stop relying on the $\pm 50\%$ day-over-day rule.

On 2026-07-30 Shopping recorded 6.0 conversions against an expected 22.33, which was a real tracking outage. On 2026-08-05 Retargeting spent 300.59 USD against an expected 169.61 USD, a real spend spike. On 2026-08-01 Brand Search recorded 6.0 conversions against an expected 0.99, a real conversion surge. The 2026-08-02 Shopping flag (14.0 against 27.24 expected) was a false alarm, so treat it as lower priority. On a real account, flags prompt an investigation and are not findings.

Measurement & follow-up

Expected effect

Catching tracking outages and spend spikes early protects the conversion data the budget moves depend on.

Measurement

Count alerts and confirmed problems each week. TabPFN had 1 false alarm among 4 alerts, against 22 among 24 for the $\pm 50\%$ rule.

Reversal

Alert method changes are reversible. If TabPFN misses real problems, bring back the $\pm 50\%$ rule alongside it.

04 A change, approved and verified

sample account demo-google

Nothing reaches an ad account without a person's approval: the reviewer sees a summary written from the stored proposal, approves it, and the change is applied once and read back.

Proposed change for review (demo-google, google_ads, g-103):

- daily_budget: 300.0 account currency per day -> 240 account currency per day
- Risk: medium (budget_delta)
- Reason: Performance Max spend fell while CPA rose; reduce daily budget by a reversible step.
- Measurement: Compare CPA and conversions after 7 complete days at the new budget.
- Reversal: Restore the previous daily budget through a new proposal.
- Nothing changes until a reviewer approves it.

After approval: Receipt for proposal 13c97ac0-b0ae-47f4-9bfc-2f9204063eaf revision 1: status=verified, mutation_attempted=True, readback_attempts=1, verified_state=[daily_budget=240.0], reason=readback matched the approved change

Sources & data notes

Accounts: sim-northwind · 1 platform source(s) with data

Change colors show increases and decreases, not whether a result is favorable.

- all sources complete and reconciled

SOURCES

Generated Oct 2, 2026 at 17:05 UTC · Report demo_report

Analysis artifact art_f5d94639cfd7a087 · Sources art_291f04581ac85105

Analysis compute/1.0.0 (period-comparison/1) · Report schema report/1